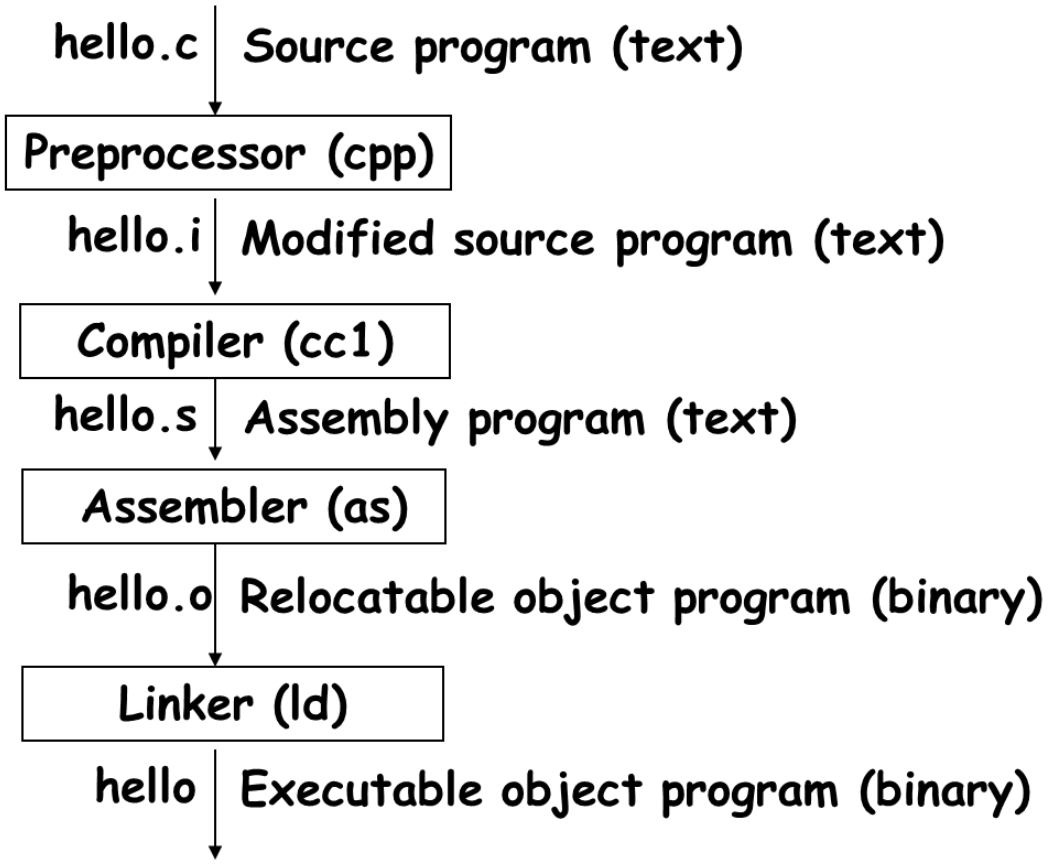
compiler

```
gcc -E test.c > test-pp.c (test-pp.c)
gcc -S test.c (test.s)
gcc -c test.c (test.o)
gcc -o test test.c  (test)
```

make

```
CC := gcc
OPT := -O3
CFLAGS := -Wall -I ./include $(OPT)

BINS := hello-1 hello-2 hello-3

all: $(BINS)

hello-1.o: hello-1.c
        gcc -c hello-1.c -Wall -O3

hello-2.o: hello-2.c
        $(CC) -c hello-2.c $(CFLAGS)

hello-3.o: hello-3.c
        $(CC) -c $^ $(CFLAGS)

clean:
        rm -f $(BINS) *.o
```

bit



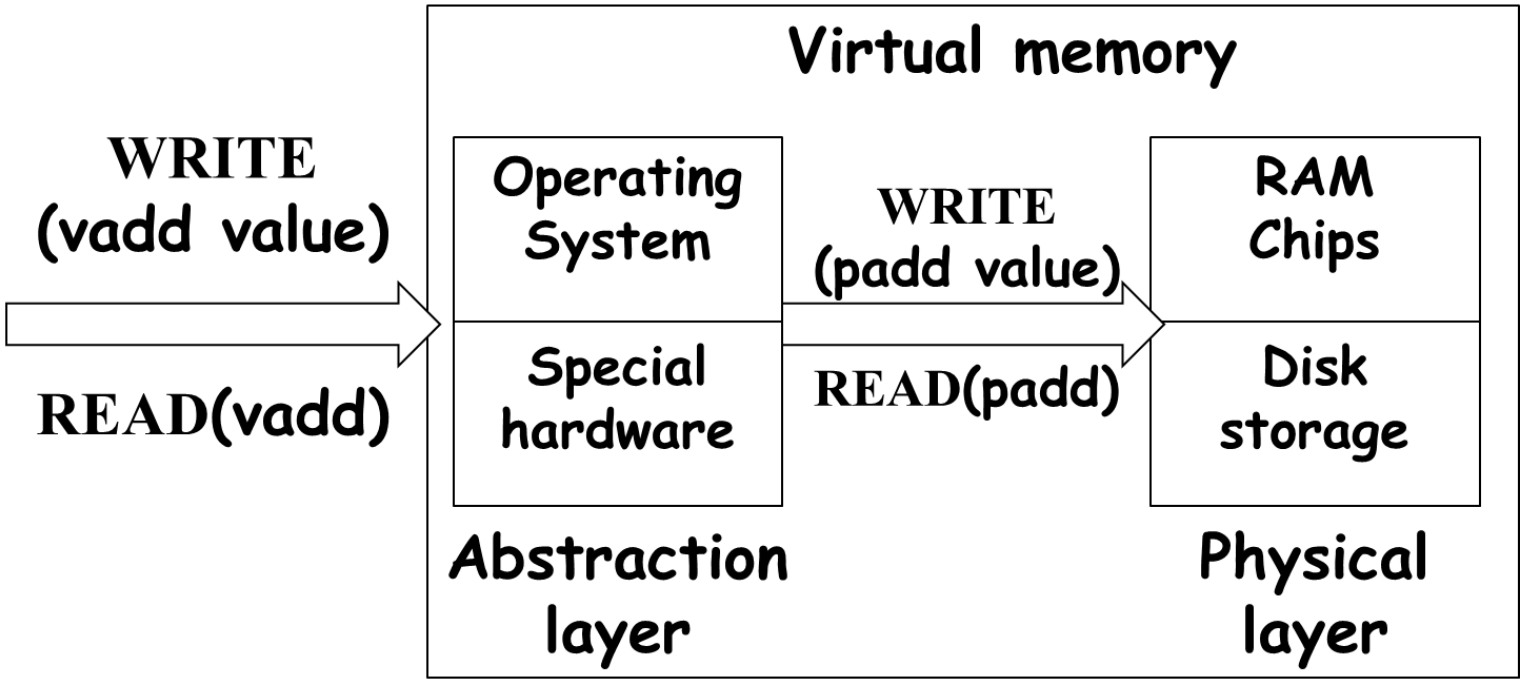
storage

data size

C Declaration		Bytes	
Signed	Unsigned	32-bit	64-bit
char	unsigned char	1	1
short	unsigned short	2	2
int	unsigned	4	4
long	unsigned long	4	8
int32_t	uint_32	4	4
int64_t	uint_64	8	8
char *		4	8
float		4	4
double		8	8

virtual memory

- Hardware
  - random-access memory (RAM) (physical)
  - disk storage (physical)
  - special hardware (performing the abstraction)
- Software
  - and operating system software (abstraction)



– 80483bd: 01 05 64 94 04 08->add %eax, 0x8049464

byte ordering

Big Endian (0x1234567)

0x100	0x101	0x102	0x103				
		01	23	45	67		

Little Endian (0x1234567)

0x100	0x101	0x102	0x103				
		67	45	23	01		

encoding

- Bit vector $[x_{w-1}, x_{w-2}, x_{w-3}, \ldots x_0]$

$$B2U(X) = \sum_{i=0}^{w-1} x_i \cdot 2^i$$

twos-complement

$$B2T(X) = -x_{w-1} \cdot 2^{w-1} + \sum_{i=0}^{w-2} x_i \cdot 2^i$$

正数的范围是0到2^(w-1)-1，而负数的范围是-1到-2^(w-1)。

unsigned和singed比较，统一转成unsigned

T2U(-2147483647) = 2^32-2147483647 = 2147483648

Constant1	Constant2	Relation	Evaluation
0	0U	==	unsigned
-1	0	<	signed
-1	0U	>	unsigned
2147483647	-2147483648	>	signed
2147483647U	-2147483648	<	unsigned
-1	-2	>	signed
(unsigned)-1	-2	>	unsigned

size_t是unsigned

```
int strlonger(char *s, char *t) {
    return strlen(s) - strlen(t) > 0;
}
```

operation

multiplication

两个w位的bit-vector相乘的结果的前w位相同

- Unsigned
  - $x *_w^u y = (x * y) \bmod 2^w$
- Signed
  - $x *_w^t y = U2T((x * y) \bmod 2^w)$
- Given two bit vectors $\vec{x}$ and $\vec{y}$
  - $x = B2T_w(\vec{x}), y = B2T_w(\vec{y}), \quad x' = B2U_w(\vec{x}), y' = B2U_w(\vec{y})$
- $T2B_w(x *_w^t y) = U2B_w(x' *_w^u y')$
  - $x' = x + x_{w-1}2^w, \quad y' = y + y_{w-1}2^w \quad (x' * y') \bmod 2^w = (x * y) \bmod 2^w$
  - $T2U_w(x *_w^t y) = T2U_w(U2T((x * y) \bmod 2^w) = (x * y) \bmod 2^w = x' *_w^u y'$

$$T2B_w(x *_w^t y) = UTB_w(T2U_w(x *_w^t y)) = UTB_w(x' *_w^t y')$$

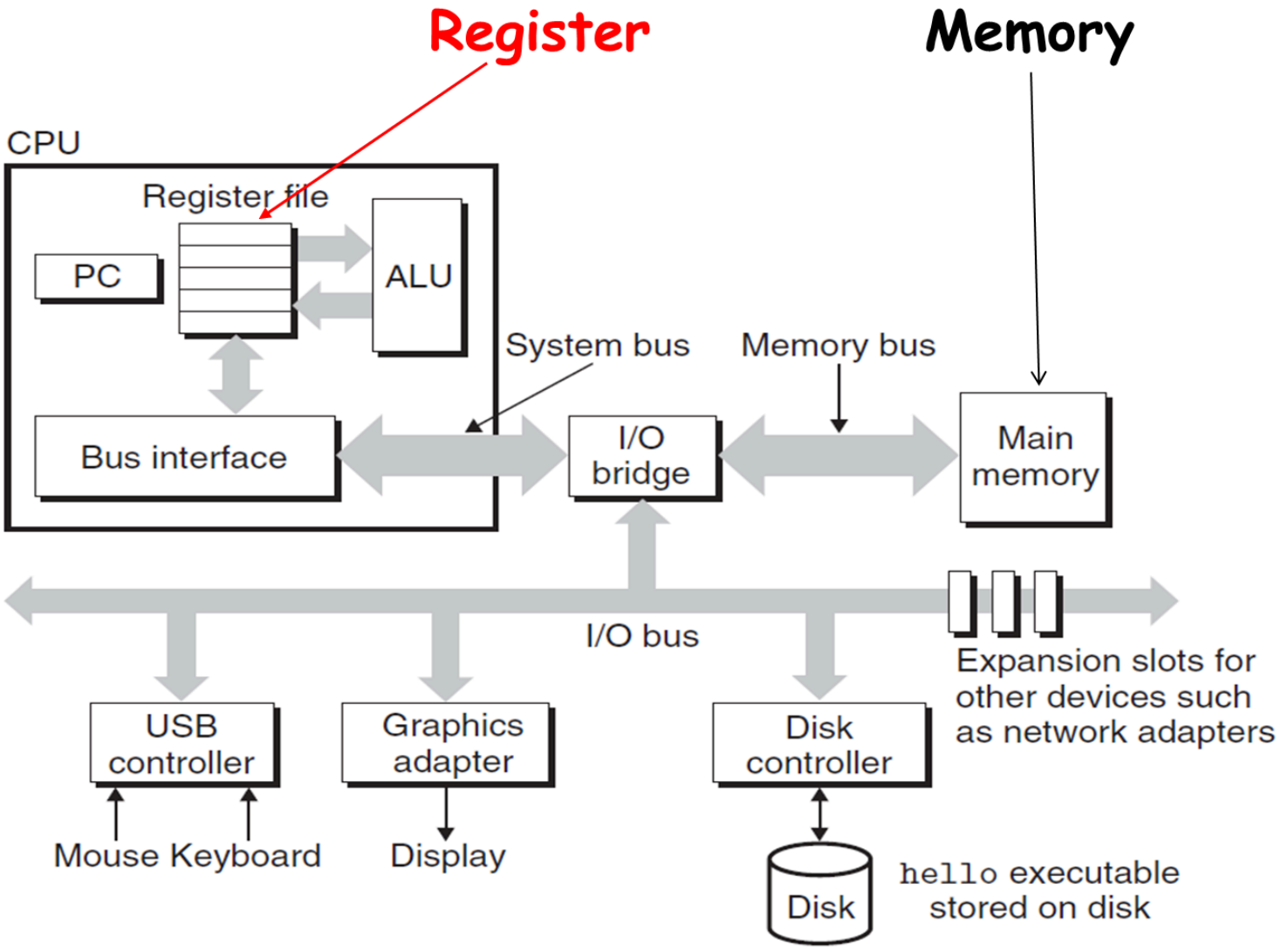
Mode	x		y		$x \cdot y$		Truncated $x \cdot y$	
Unsigned	5	[101]	3	[011]	15	[001111]	7	[111]
Two's complement	−3	[101]	3	[011]	−9	[110111]	−1	[111]
Unsigned	4	[100]	7	[111]	28	[011100]	4	[100]
Two's complement	−4	[100]	−1	[111]	4	[000100]	−4	[100]
Unsigned	3	[011]	3	[011]	9	[001001]	1	[001]
Two's complement	3	[011]	3	[011]	9	[001001]	1	[001]

4-bit	(x + y)	(x - y)	(x * y)	(-y)
(x = 4, y = 7)	-5	-3	-4	-7
(x = -6, y = -8)	2	2	0	-8
(x = 5, y = -1)	4	6	-5	1
(x = -3, y = 6)	3	7	-2	-6

-3 * 6解法

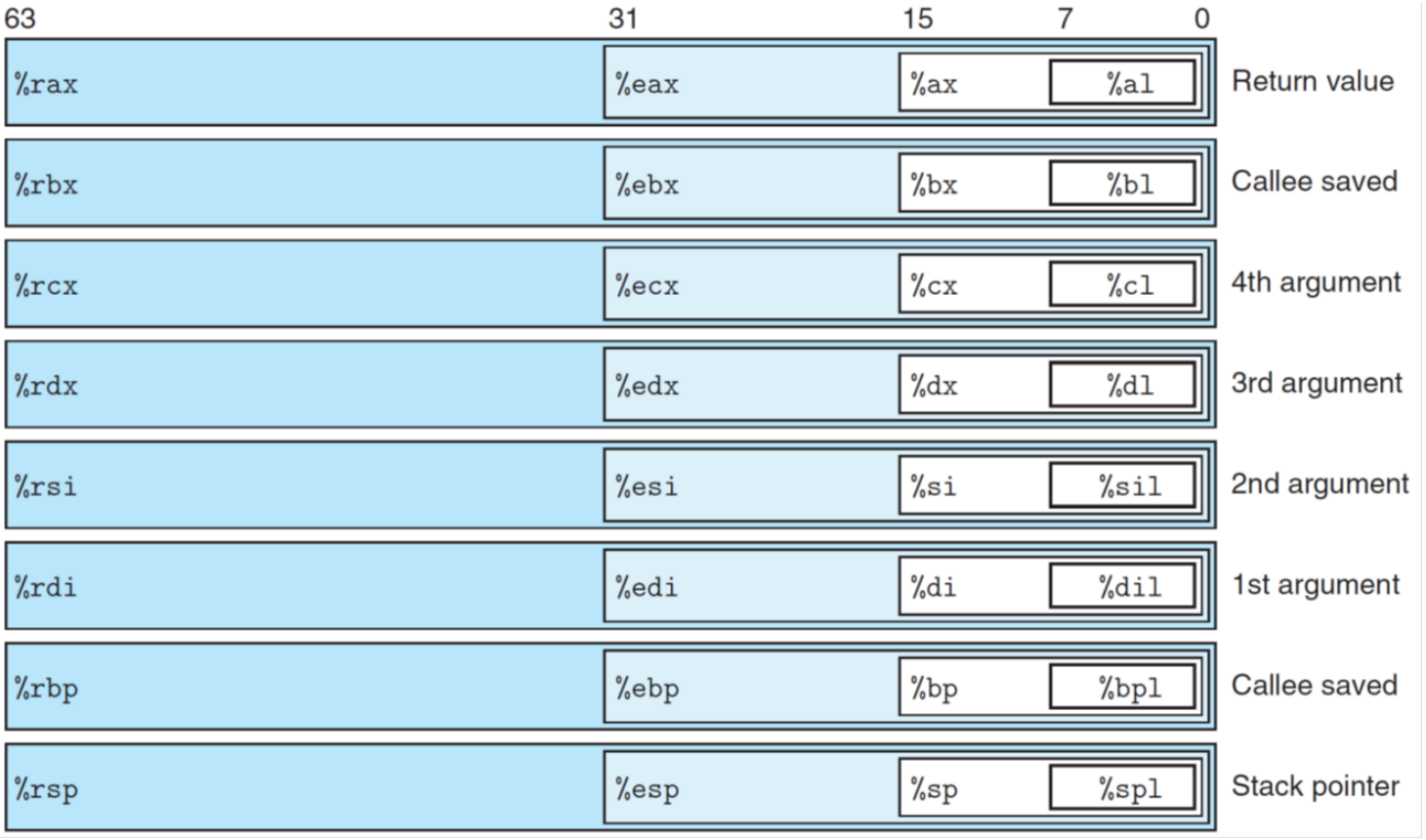
1. -3 *6 = U2T(13* 6 mod 16) = U2T(14) = 14 - 16 = -2
2. -3 * 6 = -18 = ~(0b10010) + 1 = 0b01110
   trunc(0b01110) = 0b1110 = -2

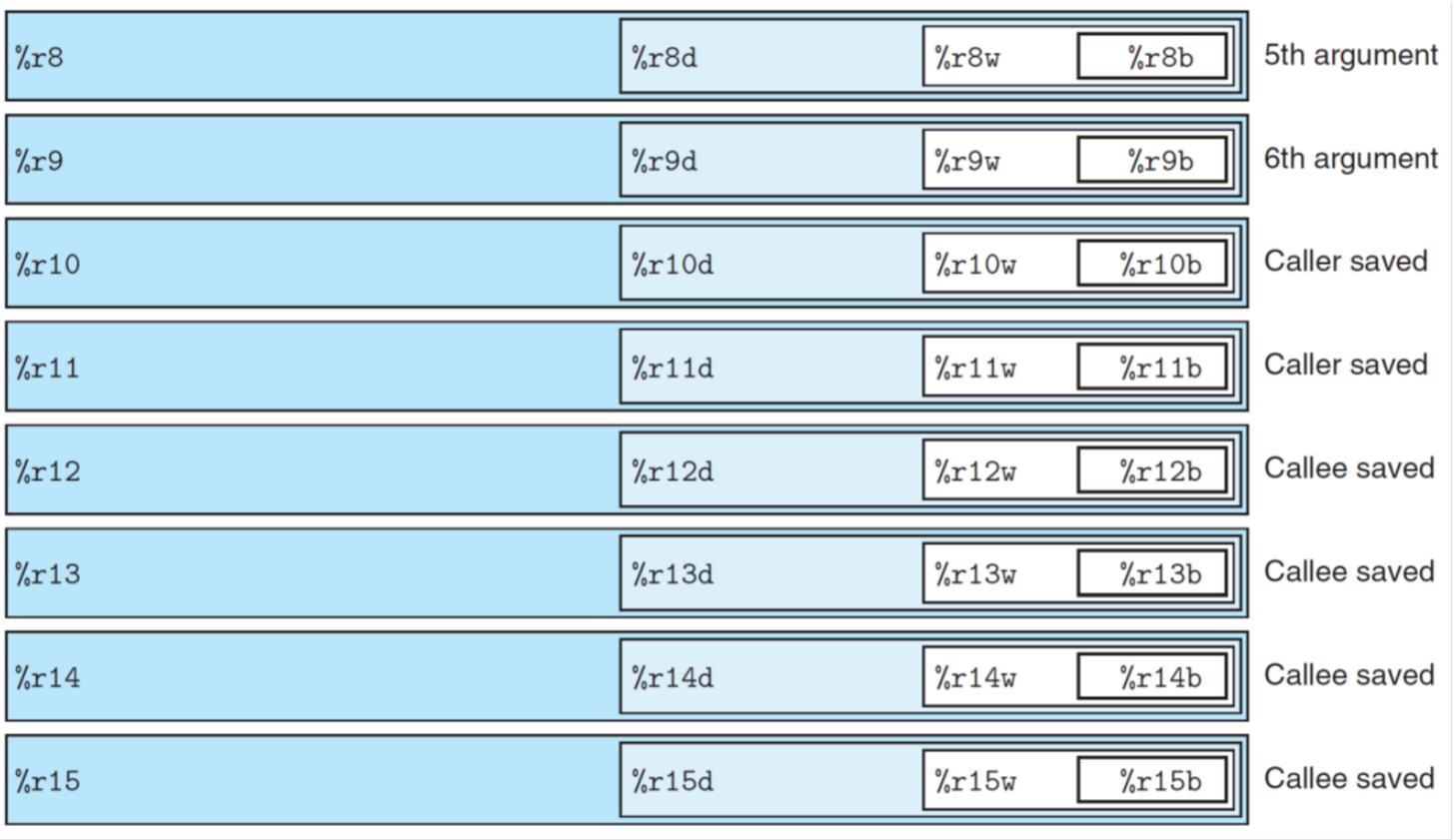
Machine-Level Representation of Programs



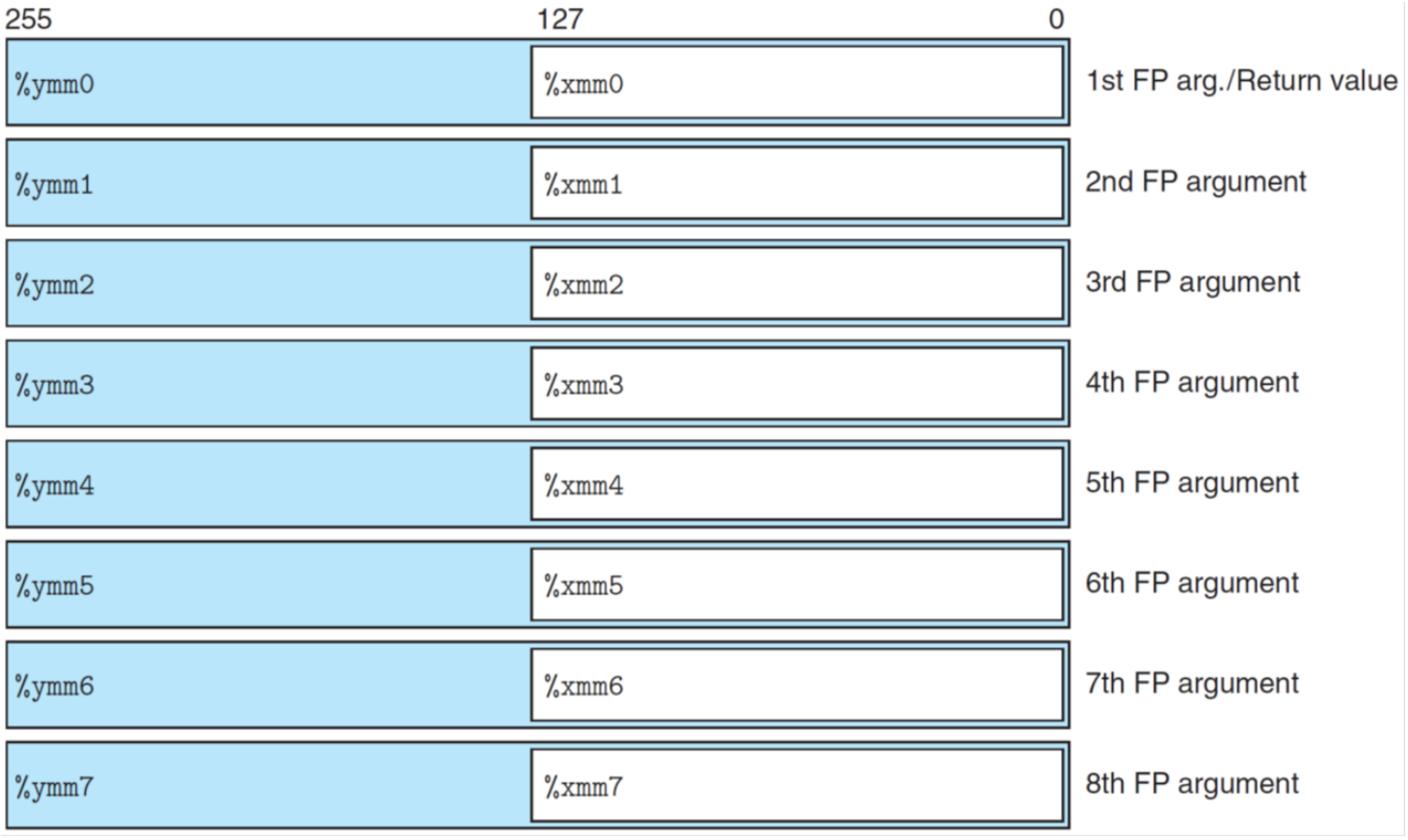
registrer

16 8-byte integer registers





16 floating point registers



%ymm8	%xmm8	Caller saved
%ymm9	%xmm9	Caller saved
%ymm10	%xmm10	Caller saved
%ymm11	%xmm11	Caller saved
%ymm12	%xmm12	Caller saved
%ymm13	%ymm13	Caller saved
%ymm14	%xmm14	Caller saved
%ymm15	%xmm15	Caller saved

address

- $Imm(r_b, r_i, s)$

```
function(%rdi, %rsi, %rdx)
    return %rax
```

Data movement

- (immediate, register)
- (memory, register) load
- (register, register)
- (immediate, memory) limited
- (register, memory) store

- w 字(2 bytes) 16位
- d,l 双字 32位
- q 四字 64位

- mov (general)
- movb (move byte)
- movw (move word)
- movl (move double word)
- movq (move quad word)
  - Immediate can only be represented as 32-bit signed sign extension to the destination
- movabsq (move absolute quad word)

- Only update the specific register byte
  - except movl which updates the higher-order 4 bytes of the register to 0

unsigned

- movz (move with zero extension)
- movzbw (move zero-extension byte to word)
- movzbl (move zero-extension byte to double word)
- movzbq (move zero-extension byte to quad word)
- movzwl (move zero-extension word to double word)
- movzwq (move zero-extension word to quad word)
- no movzlq, it is implemented by movl

signed

- movs (move with sign extension)
- movsbw (move sign-extension byte to word)
- movsbl (move sign-extension byte to double word)
- movsbq (move sign-extension byte to quad word)
- movswl (move sign-extension word to double word)
- movswq (move sign-extension word to quad word)
- movslq (move sign-extension double word to quad word)

example

Initial value %dl=8d %rax =0000000098765432

1	movb	%dl, %al	%rax=0000000098765 48d
2	movsbl	%dl, %eax	%rax=00000000ffffff8d
3	movzbq	%dl, %rax	%rax=000000000008d

Stack

%rsp register stack pointer

```
int main()
{
        int a = 4 , b = 3, c = 2;
        decode1(&a, &b, &c);
        printf("a = %d, b = %d, c = %d\n", a, b, c);
}
```

Data manipulation

注意

- sub S, D 是 D <- D - S被减数和存入的都在右侧
- sar算术右移，shr逻辑右移
- idivq S右边有长整数除法

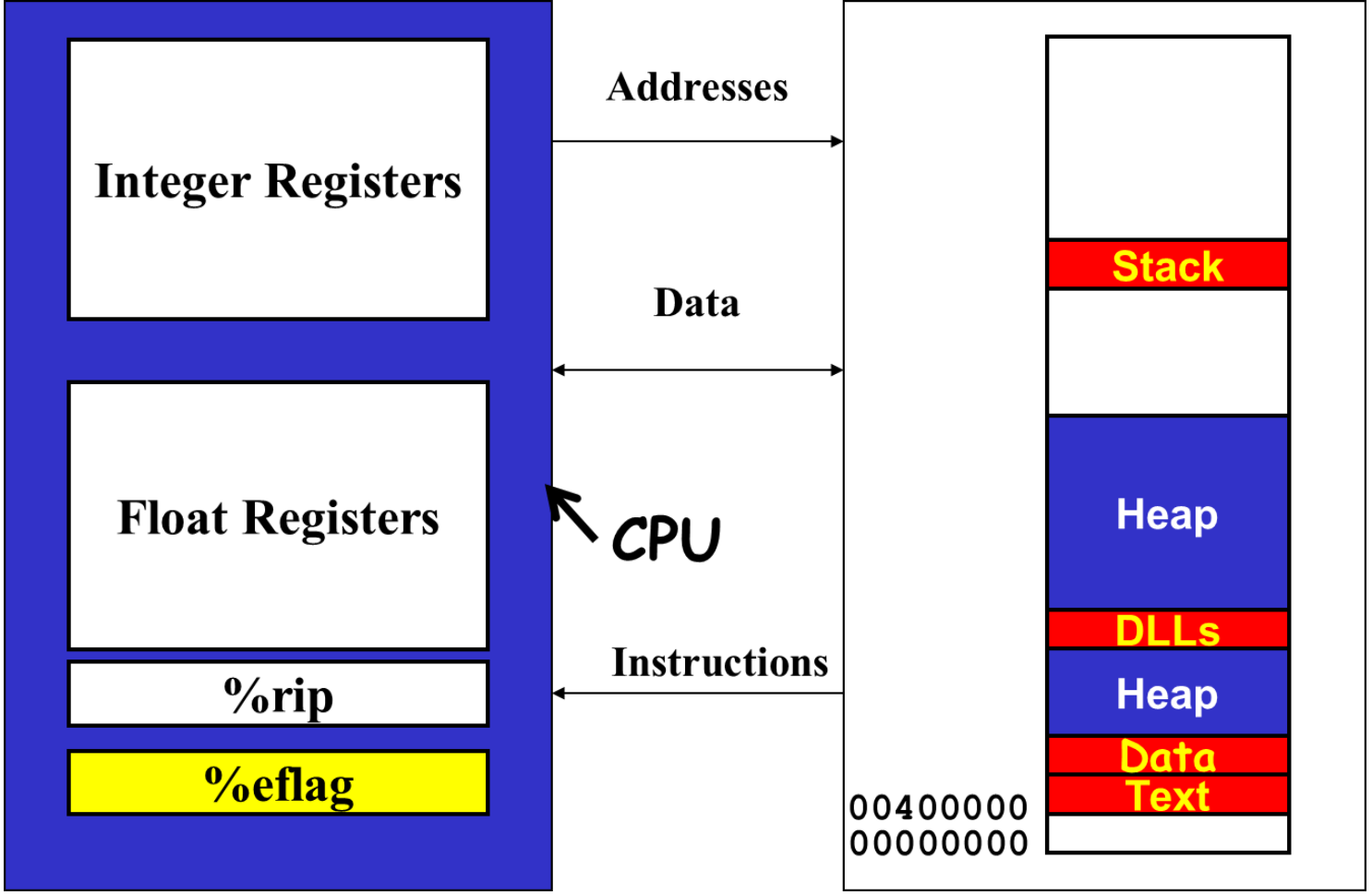
Instruction	Effect	Description
leaq S, D	D ← &S	Load effective address
inc D	D ← D + 1	Increment
dec D	D ← D – 1	Decrement
neg D	D ← -D	Negate
not D	D ← ~D	Complement
add S, D	D ← D + S	Add
sub S, D	D ← D – S	Subtract
imul S, D	D ← D * S	Multiply

Instruction	Effect	Description
xor S, D	D ← D ^ S	Exclusive-or
or S, D	D ← D | S	Or
and S, D	D ← D & S	And
sal k, D	D ← D << k	Left shift
shl k, D	D ← D << k	Left shift
sar k, D	D ← D >> k	Arithmetic right shift
shr k, D	D ← D >> k	Logical right shift

imulq S	R[%rdx]:R[%rax] ←S*R[%rax]	Signed full multiply
mulq S	R[%rdx]:R[%rax] ←S*R[%rax]	Unsigned full multiply
cqto	R[%rdx]:R[%rax] ← SignExtend(R[%rax])	Convert to oct word
idivq S	R[%rdx] ← R[%rdx]:R[%rax] mod S R[%rax] ← R[%rdx]:R[%rax] ÷ S	Signed divide
divq S	R[%rdx] ←R[%rdx]:R[%rax] mod S R[%rax] ← R[%rdx]:R[%rax] ÷ S	Unsigned divide

control

%rip PC 64位
%eflag 32位



change CD

%eflag 存condition codes 32位

- CF: carry flag
- OF: overflow flag
- ZF: zero flag
- SF: sign flag

- lea instruction
  - has no effect on condition codes
- Xorl instruction
  - The carry and overflow flags are set to 0
- Shift instruction
  - carry flag is set to the last bit shifted out
  - Overflow flag is set to 0

非算术运算符修改CD

- `test b,a` like computing `a&b` without setting destination
- ZF set when `a&b == 0`
- SF set when `a&b < 0`

指令		基于	描述
CMP	S_1，S_2	S_2-S_1	比较
cmpb			比较字节
cmpw			比较字
cmpl			比较双字
cmpq			比较四字
TEST	S_1，S_2	$S_1 \& S_2$	测试
testb			测试字节
testw			测试字
testl			测试双字
testq			测试四字

- **cmp b,a** like computing a-b without setting destination
- CF set if carry out from most significant bit
  - Used for unsigned comparisons
- ZF set if a == b
- SF set if (a-b) < 0
- OF set if two's complement overflow

```
(a>0 && b<0 && (a-b)<0) ||
(a<0 && b>0 && (a-b)>0)
```

read CD

Instruction	Synonym	Effect	Set Condition
Sete	Setz	ZF	Equal/zero
Setne	Setnz	~ZF	Not equal/not zero
Sets		SF	Negative
Setns		~SF	Nonnegative
Setl	Setnge	SF^OF	Less
Setle	Setng	(SF^OF)|ZF	Less or Equal
Setg	Setnle	~(SF^OF)&~ZF	Greater
Setge	Setnl	~(SF^OF)	Greater or Equal
Seta	Setnbe	~CF&~ZF	Above
Setae	Setnb	~CF	Above or equal
Setb	Setnae	CF	Below
Setbe	Setna	CF|ZF	Below or equal

jmp

rep作为一个优化，不执行动作。

```
loop:
1   4004d0:   48 89 f8movq   %rdi, %rax
2   4004d3:   eb 03          jmp    8<loop+0x8>
3   4004d5:   48 di f8 sari  %rax
4   4004d8:   48 85 c0       testq  %rax, %rax
5   4004db:   7f f8          jq     5<loop+0x5>
6   4004dd:   f3 c3          repz retq
```

03 + 4004d5 = 4004d8

f8(-8) + 4004dd = 4004d5

如果有lt_cnt和ge_cnt

```
long lt_cnt = 0 ;
long ge_cnt = 0 ;
long absdiff_se(long x, long y)
{
    long result ;
    if (x < y) {
        lt_cnt++
        return y – x;
    }
    else {
        ge_cnt++ ;
        return x – y;
    }
}
```

```
1.  long gotodiff_se(long x, long y)
2.  {
3.      long result ;
4.      if (x >= y)  goto x_ge_y ;
5.      lt_cnt++ ;
6.      result = y – x ;
7.      return result ;
8.   x_ge_y:
9.      ge_cnt++ ;
10.     result = x – y;
11.     return result;
12. }
```

14

(a) Original C code

```
long absdiff(long x, long y)
{
    long result ;
    if (x < y)
        restul = y-x ;
    else
        result = x-y ;
    return result ;
}
```

(b) Implementation using conditional assignment

```
1 long cmovdiff(long x, long y) {
2     long rval = y-x;
3     long eval = x-y;
4     long ntest = x >= y;
5     /* Line below requires
6        single instruction: */
7     if (ntest) rval = eval;
8     return rval;
9 }
```

16

conditional move

源寄存器或内存地址S，目的寄存器R

Instruction		Synonym	Move condition	Description
cmove	S, R	cmovz	ZF	Equal / zero
cmovne	S, R	cmovnz	~ZF	Not equal / not zero
cmovs	S, R		SF	Negative
cmovns	S, R		~SF	Nonnegative
cmovg	S, R	cmovnle	~(SF ^ OF) & ~ZF	Greater (signed >)
cmovge	S, R	cmovnl	~(SF ^ OF)	Greater or equal (signed >=)
cmovl	S, R	cmovnge	SF ^ OF	Less (signed <)
cmovle	S, R	cmovng	(SF ^ OF) | ZF	Less or equal (signed <=)
cmova	S, R	cmovnbe	~CF & ~ZF	Above (unsigned >)
cmovae	S, R	cmovnb	~CF	Above or equal (Unsigned >=)
cmovb	S, R	cmovnae	CF	Below (unsigned <)
cmovbe	S, R	cmovna	CF | ZF	below or equal (unsigned <=)

destination must be register. source doesn't matter

不能用cmove，*xp始终会执行

```
int cread(int *xp) {
    return (xp ? *xp : 0);
}
```

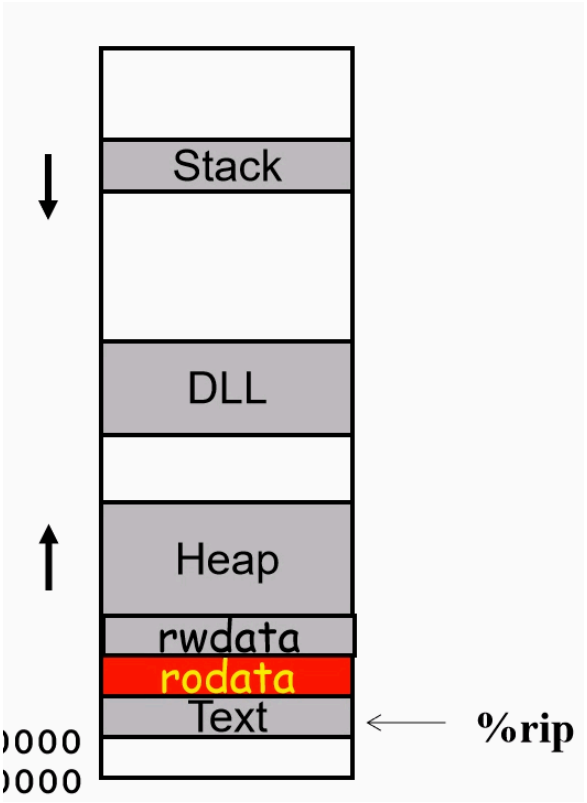
when are the use of registers not saved

- Leave procedures
- Inter-procedure register allocation
- a1 is dead before f calls h()
- Register window

jmp table

```
int switch_eg_impl (long x,
            long n, long *dest)
{
   static void *jt[7] = { &&loc_A,
              &&loc_def, &&loc_B,
              &&loc_C,   &&loc_D,
              &&loc_def, &&loc_D
             };

   unsigned long index = n - 100;
   long val ;
   if ( index > 6 )
         goto loc_def ; //default
   goto *jt[index];     // multiway
loc_A:  // case 100
   val = x * 13 ;
   goto done ;
```

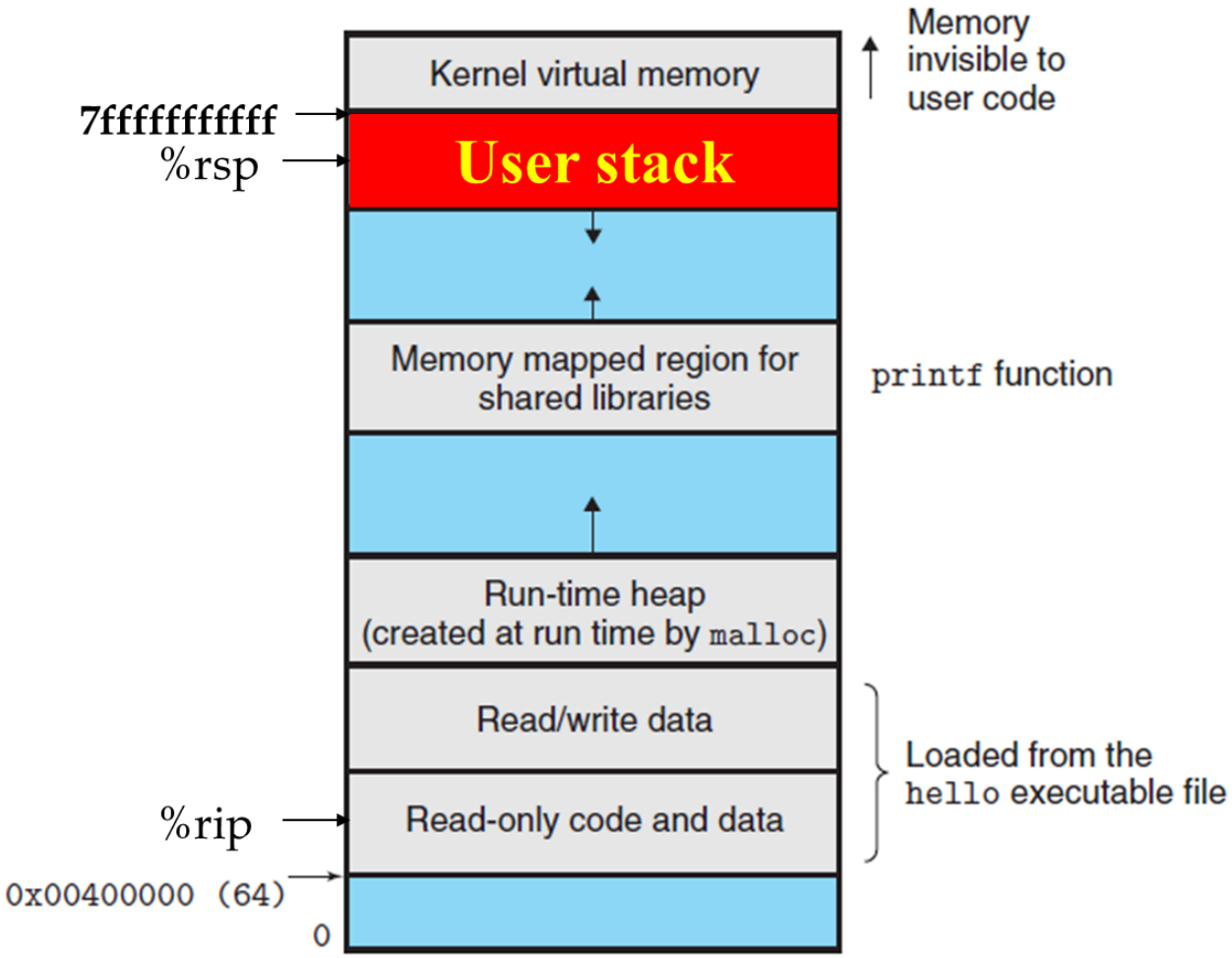
```
loc_B:   // case 102
   x = x + 10 ;  // fall through
loc_C: // case 103
   val = x + 11;
   goto done ;
loc_D: // case 104, 106
   val *= x * x ;
   goto done ;
loc_def:  //default
   val = 0 ;
done:
   *dest = val ;
}
```



procedure call

jmp with: return, pass data, local variable, register

caller: 调用者 callee: 被调用者

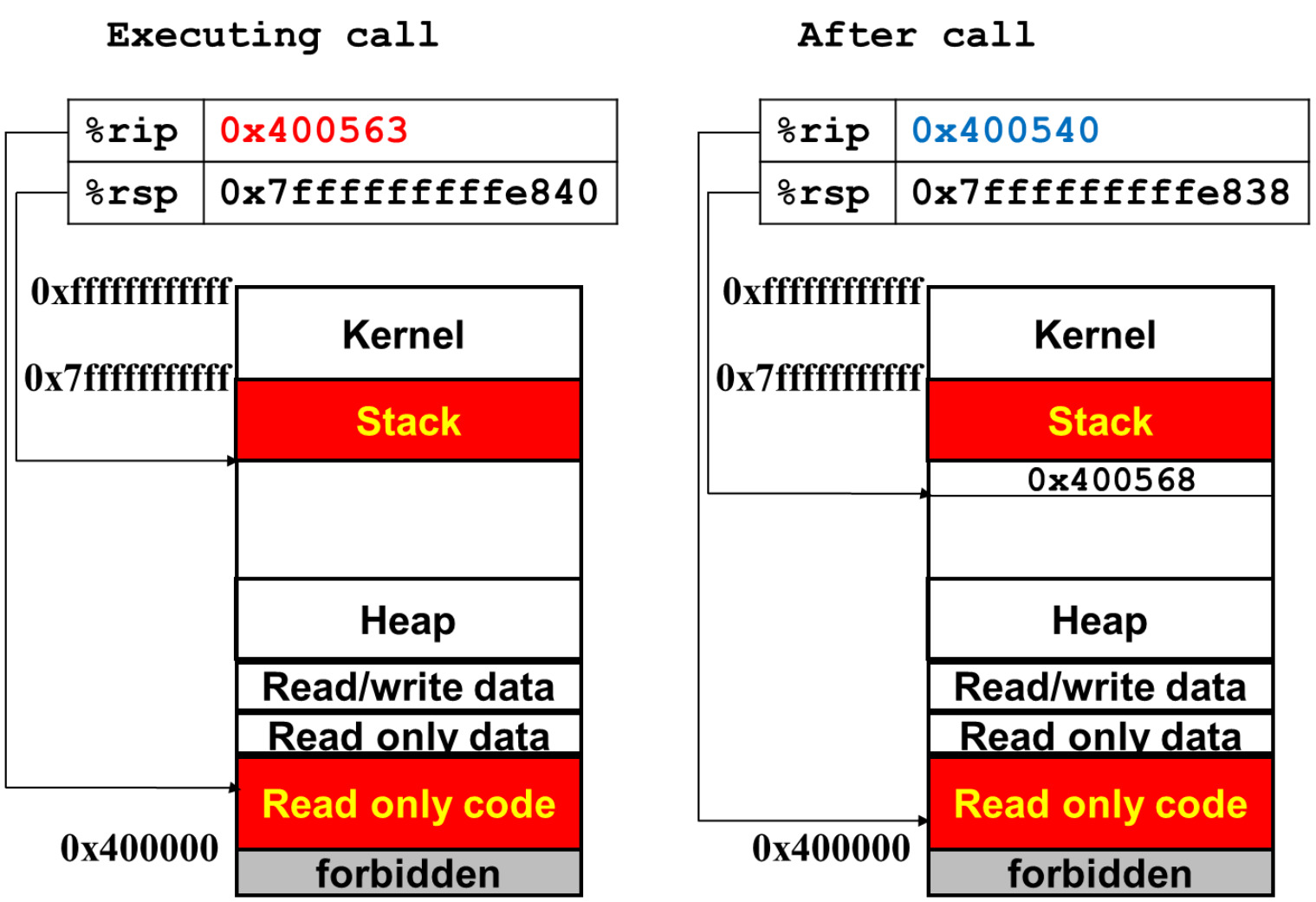


- call = push + jmp
- ret = pop + jmp

example

```
    // beginning of function multstore
1.  400540 <multstore>:
2.  400540: 53                  push   %rbx
3.  400541: 48 89 d3            mov    %rdx, %rbx
    . . .
    // return from function multstore
4.  40054d: c3                  retq
    . . .
```

```
    //call to mulstore from main
5.  400563: e8 d8 ff ff ff      callq  400540<multstore>
6.  400568: 48 8b 54 24 08      mov    0x8(%rsp), %rdx
```



Operand size	Argument number					
	1	2	3	4	5	6
64	%rdi	%rsi	%rdx	%rcx	%r8	%r9
32	%edi	%esi	%edx	%ecx	%r8d	%r9d
16	%di	%si	%dx	%cx	%r8w	%r9w
8	%dil	%sil	%dl	%cl	%r8b	%r9b

多于6个参数

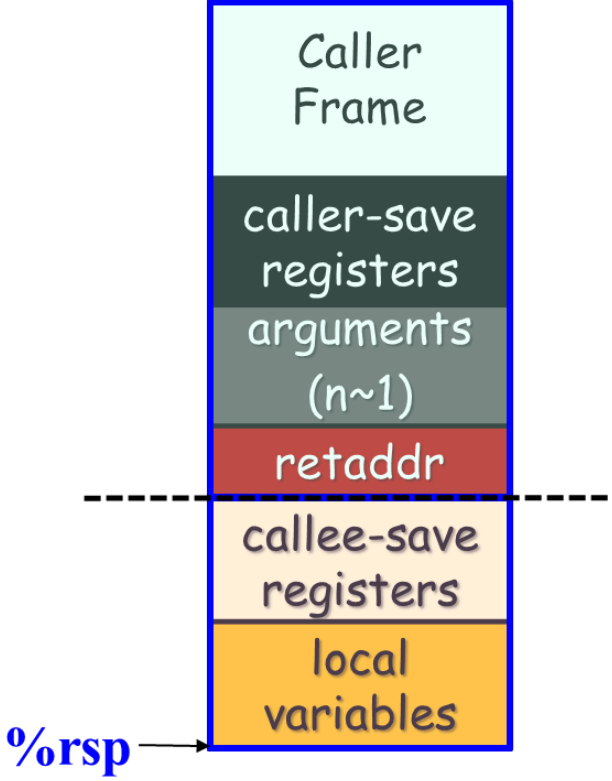
- Pushed by Caller to stack
  - Saved in caller frame
  - Just upon of return address
  - From N^{th} to 7^{th} (from right to left)
  - All data sizes are rounded up to be multiples of eight
- How to be used by Callee
  - Relative to %rsp

caller-save callee可以随便修改，caller保存才能恢复

- Caller-save registers
  - %rax, %rdi, %rsi, %rdx, %rcx, %r8, %r9, %r10, %r11

callee-save 对于caller来说，callee不会修改这些值

- Callee-save registers
  - %rbx, %rbp, %r12-15




```
long P(long x, long y)
 {
        long u = Q(y);
        long v = Q(x);
        return u + v;
 }
```

```
1    P:
2        pushq   %rbp          Save %rbp
3        pushq   %rbx          Save %rbx
4        subq    $8, %rsp      Align stack frame
5        movq    %rdi, %rbp    Save x
6        movq    %rsi, %rdi    Move y to first argument
7        call Q  Call Q(y)
8        movq    %rax, %rbx    Save result
9        movq    %rbp, %rdi    Move x to first argument
10       call Q  Call Q(x)
11       addq    %rbx, %rax    Add saved Q(y) to Q(x)
12       addq    $8, %rsp      Deallocate last part of stack
13       popq    %rbx          Restore %rbx
14       popq    %rbp          Restore %rbp
15       ret
```

array

```
int var_ele(long n, int A[n][n], long i, long j)
  {
    return A[i][j];
  }
```

- Declare an array int A[exp1][exp2]
  - either as a local variable
  - or as an argument to a function

Heterogeneous Data Structures

struct & union

```
struct S3 {
   char c;
   int i[2];
   double v;
};
union U3 {
   char c;
   int i[2];
   double v;
};
```

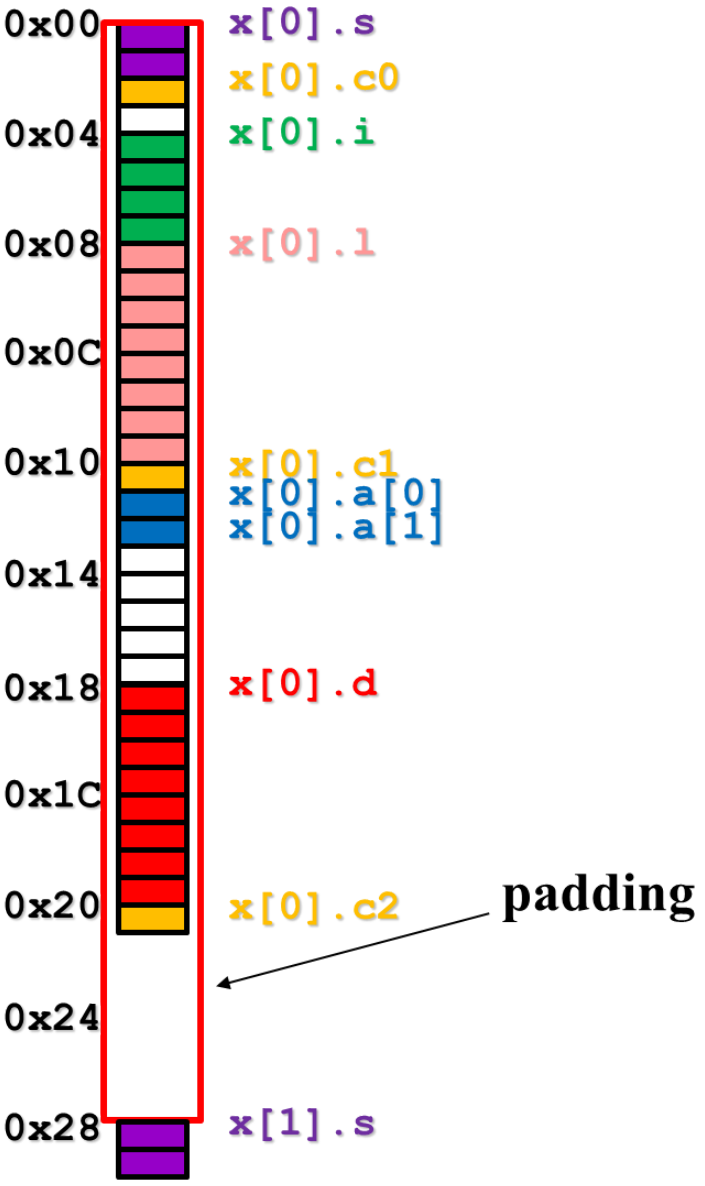
Type	c	i	v	size
S3	0	4	16	24
U3	0	0	0	8

align

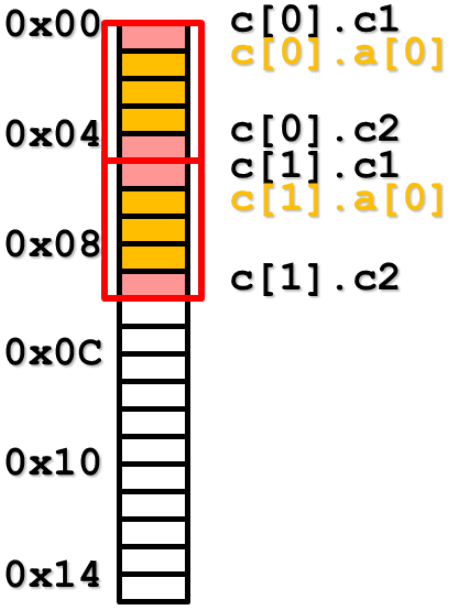
- Linux alignment restriction
  - 1-byte data types are able to have any address
  - 2-byte data types must have an address that is multiple of 2
  - 4-byte data types must have an address that is multiple of 4
  - Any larger data types must have an address that is multiple of 8

example

```
struct xxx {
     short s;
     char c0;
     int i;
     long l;
     char c1;
     char a[2];
     double d;
     char c2;
};

struct xxx x[2];
```




```
struct ccc {
     char c1;
     char a[3];
     char c2;
};

struct ccc c[2];
```



buffer overflow

prevent

1. stack randomization
2. stack corruption detect: canary

```
1 echo:
2       subq    $24, %rsp               # Allocate 24 bytes on stack
3       movq    %fs:40, %rax            # Retrieve canary
4       movq    %rax, 8(%rsp)           # Store on stack
5       xorl    %eax, %eax              # Zero out register
6       movq    %rsp, %rdi              # Compute buf as %rsp
7       call    gets                    # Call gets
8       movq    %rsp, %rdi              # Compute buf as %rsp
9       call    puts                    # Call puts
10      movq    8(%rsp), %rax           # Retrieve canary
11      xorq    %fs:40, %rax            # Compare to stored value
12      je      .L9                     # If =, goto ok
13      call    __stack_chk_fail        # Stack corrupted!
14  .L9                                 # ok:
15      addq    $24, %rsp               # Deallocate stack space
16      ret
```

3. limit excutable code region: stack not excutable

return-oriented programming

pointer

- Addition and subtraction
  - p+i , p-i (result is a pointer)
  - p-q (result is a long)

```
int numcmp(char *s1, char *s2)
{
    double v1, v2;

    v1 = atof(s1);
    v2 = atof(s2);
    if (v1 < v2) return -1;
    else if ( v1 > v2 ) return 1;
    else return 0;
}

/* strcmp: compare s1 and s2 as string */
int strcmp(char *s1, char *s2);
```

```
#include <stdio.h>
#include <string.h>

{
    ...
    int numeric = 0, (*cmp)(void *, void *);
    char *s1, *s2;
    ...
    if (...) numeric = 1 ;
    ...
    cmp = (int (*)(void *, void *))
              (numeric ? numcmp : strcmp);
    (*cmp)(s1, s2);
    ...
}
```

- char (*(*x())[])()
  - Function returning pointer to array[ ] of pointer to function returning char
- char (*(*x[3])())[5]
  - Array[3] of pointer to function returning pointer to array[5] of char

Operators	Associativity
() [] -> . ++ --	left to right
! ~ ++ -- + - * & (type) sizeof	right to left
* / %	left to right
+ -	left to right
<< >>	left to right
< <= > >=	left to right
== !=	left to right
&	left to right
^	left to right
|	left to right
&&	left to right
||	left to right
?:	right to left
= += -= *= /= %= &= ^= != <<= >>=	right to left
,	left to right

Note: Unary +, -, and * have higher precedence than binary forms